

Math 2551 Worksheet: Surfaces

1. Consider the surface cut from the parabolic cylinder $y = 4 - x^2$ by the planes $z = 0$, $z = 2$, and $y = 0$. Sketch S and find a parameterization of S .
2. Consider the surface left in the hemisphere $x^2 + y^2 + z^2 = 4$, $z \geq 0$ after cutting off the lower part between the planes $z = 0$ and $z = \sqrt{3}$ from the hemisphere. (In short, S is the surface which is the part of $x^2 + y^2 + z^2 = 4$ above $z = \sqrt{3}$). Sketch S , parametrize S and find the surface area of S .
3. The tangent plane at a point $P_0 = (f(u_0, v_0), g(u_0, v_0), h(u_0, v_0))$ on a parameterized surface $\mathbf{r}(u, v) = \langle f(u, v), g(u, v), h(u, v) \rangle$ is the plane through P_0 normal to the vector $\mathbf{r}_u(u_0, v_0) \times \mathbf{r}_v(u_0, v_0)$.

Use this to find an equation to the tangent plane of the surface parameterized by $\mathbf{r}(r, \theta) = (r \cos(\theta))\mathbf{i} + (r \sin(\theta))\mathbf{j} + r\mathbf{k}$, $r \geq 0$, $0 \leq \theta \leq 2\pi$ at the point where $(r, \theta) = (2, \pi/4)$.

What is a Cartesian equation for this surface? Sketch it and the tangent plane.

4. Find the area of the part of the surface $z = xy$ that lies within the cylinder $x^2 + y^2 = 1$.
5. Find the area of the surface cut from the “nose” of the paraboloid $x = 1 - y^2 - z^2$ by the yz -plane.